Programming for Finance

Accessing Financial Data via APIs

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Signs

– Please take pictures of the information. Medium priority.

Source: https://www.flaticon.com/ru/free–icon/camera–flash_3460307

– Please write down the information. High priority.

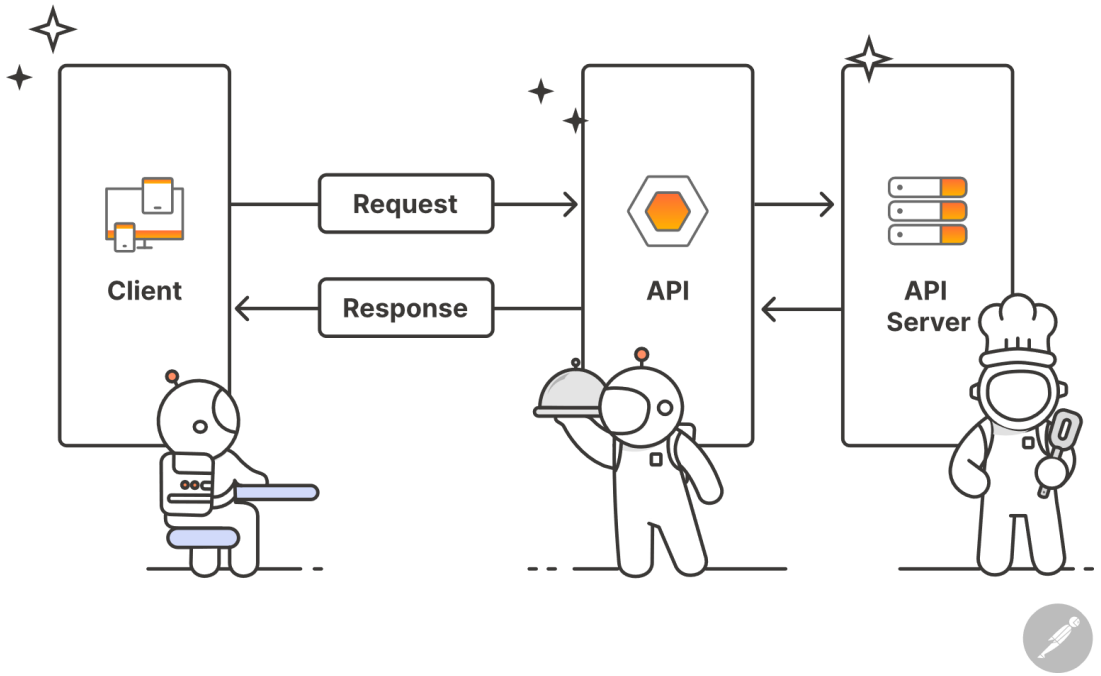
Source: https://openmoji.org/library/emoji–270D/

TOPICS

1. API
2. Yahoo Finance

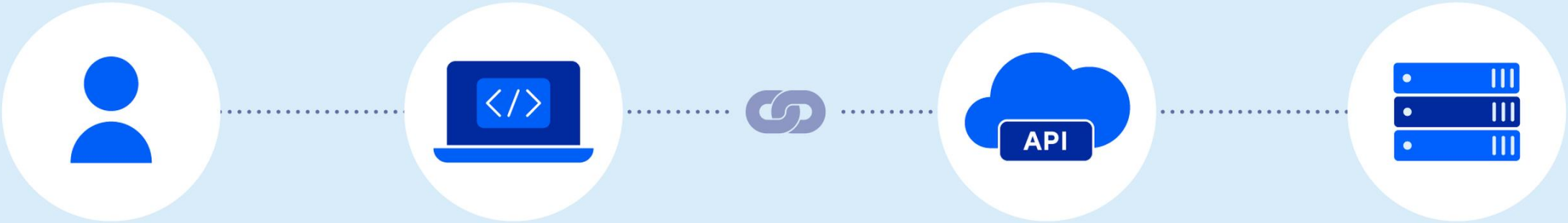
API (Application programming interface)

- An API is a standard for communicating to a server/website.
- Each application/website/server will have its own set of rules or it own unique API.
- Most APIs nowadays use JSON as their data format.



https://www.postman.com/what-is-an-api/

How does an API work?



End user

End user will get the required info on your app.

Developers

Developers will design your app in a way that it can access data stored in assets/databases via APIs.

API

API will connect with assets/databases to accessrequired info, sending it to the app.

Assets (database)

The data of the other brands will be used for providing third-party services.

https://www.adjust.com/glossary/api/

JSON Example

Attribute

Value

```
{
    "status": "ok",
    "person": {
        "name": "Bob Smith",
        "id": 800000001,
        "age": 25,
        "address": {
            "line1": "1447 Bryan Street",
            "city": "Greensboro",
            "state": "NC",
            "zip": "27407"
        }
    }
}
```

https://www.youtube.com/watch?v=LwXgnd5co4w

API Example - Yahoo Finance

- Yahoo Finance provides financial news, data and commentary including stock quotes, press releases, financial reports, and original content. It also offers online tools for personal finance management, including portfolio tracking, interactive charts and stock screening.
- Your Python Code → API Request → Financial Data Provider → Price Data → Your DataFrame

yfinance

- «Download market data from Yahoo! Finance's API»
- yfinance is not affiliated, endorsed, or vetted by Yahoo, Inc. It's an open-source tool that uses Yahoo's publicly available APIs, and is intended for research and educational purposes.
- https://pypi.org/project/yfinance/ – library description
- ranaroussi.github.io/yfinance – library documentation

Other financial data API examples

- https://massive.com/
- https://site.financialmodelingprep.com/
- https://data.nasdaq.com/